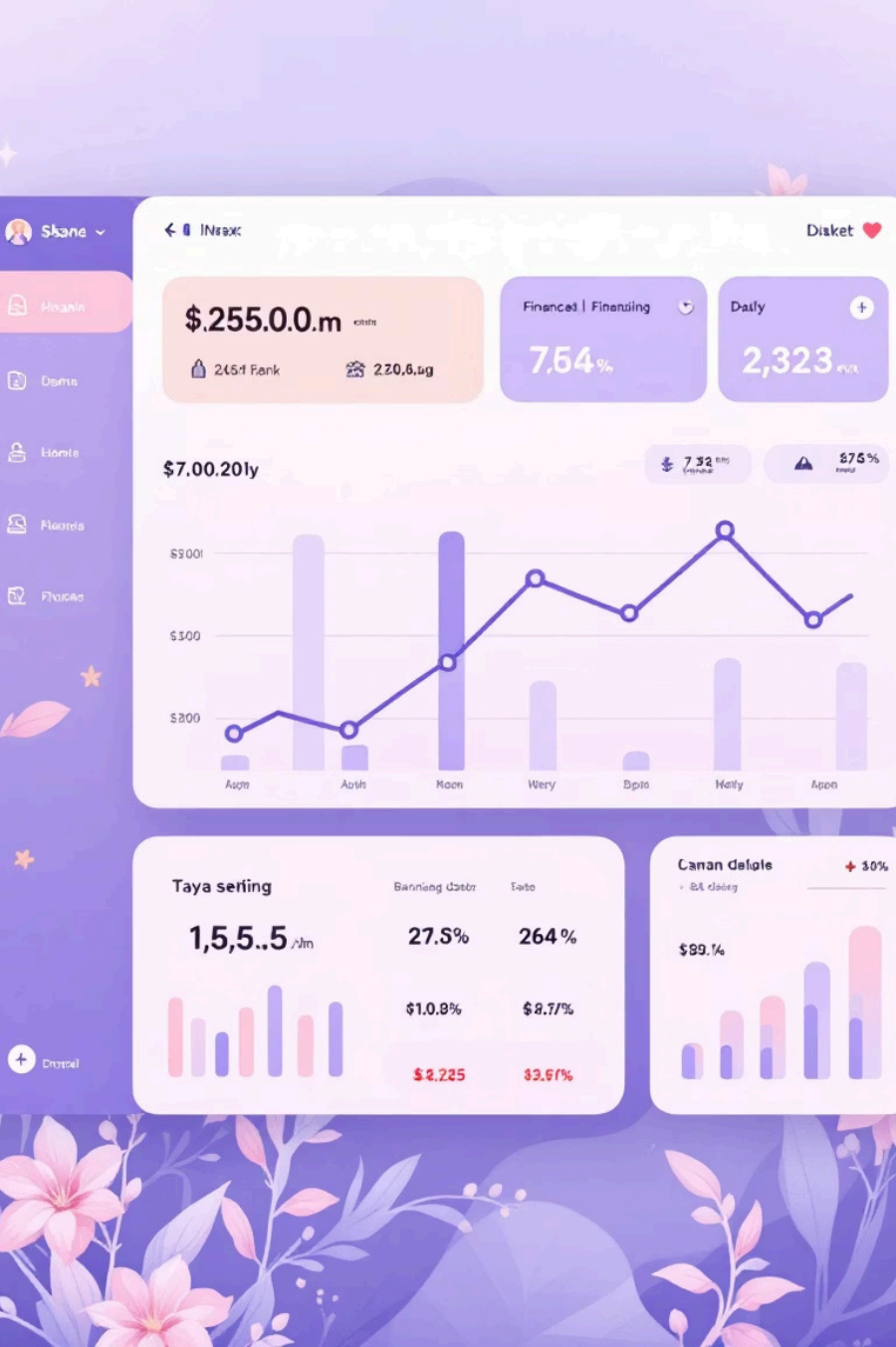




Bank Loan Analysis Project

Delivering actionable insights into loan performance, portfolio health, and risk distribution through SQL and Power BI



BUSINESS CONTEXT

Why Bank Loan Analytics Matter

Revenue Driver

Loans are critical revenue sources but carry inherent credit risk

Portfolio Health

Monitor performance and identify good vs bad loan behavior

Risk Assessment

Track DTI ratios, interest trends, and borrower creditworthiness

Data-Driven Decisions

Support strategic lending and risk management choices



Project Architecture



Data Sources

CSV files with borrower and loan data



SQL Analysis

Metric validation and aggregation logic



Power BI

Interactive dashboards with KPIs and trends



Insights

Actionable business intelligence

Data Preparation Pipeline

01

Extract

Import raw loan data from CSV into Power BI

02

Transform

Clean nulls, standardize dates, convert rates to percentages

03

Load

Build star schema with optimized relationships



- ❏ **Key transformations:** Handled inconsistent values, standardized loan status categories, enabled time intelligence calculations



Core Portfolio KPIs

5

Key Metrics

Total applications, funded amount, amount received, interest rate, DTI

MTD

Time Intelligence

Month-to-date and previous month comparison

2

Loan Categories

Good loans (Fully Paid, Current) vs Bad loans (Charged Off)

Good vs Bad Loan Classification

Good Loans

- Fully Paid status
- Current repayment
- Strong recovery performance

Bad Loans

- Charged Off status
- High credit risk
- Portfolio risk indicator

Segmentation provides immediate visibility into portfolio risk and recovery efficiency



Multi-Dimensional Analysis



Regional Analysis

Loan performance by state and geographic distribution



Monthly Trends

Time-based patterns in applications and funding



Employment Impact

Loan volume correlation with employment length



Home Ownership

Segmentation by borrower housing status



Loan Purpose

Distribution across debt consolidation, credit cards, home improvement



Loan Terms

Performance analysis by repayment duration

Power BI Dashboard Features



Executive KPI Layer

High-level cards with total applications, funded amounts, and recovery metrics

Interactive Visuals

Cross-filtered charts enabling drill-down analysis across all dimensions

Dynamic Slicers

Time-based filters for flexible MTD and PMTD comparison

Strategic Business Value

Risk Identification

Pinpoint high-risk loan segments for proactive management

Repayment Efficiency

Understand recovery patterns across borrower profiles

Regional Intelligence

Assess demand and recovery by geographic location

Portfolio Optimization

Data-backed decisions for lending strategy



"The dashboard supports both operational monitoring and strategic planning use cases"



Project Impact & Future Roadmap

1

Current State

Complete analytics lifecycle from raw data to executive insights

2

Foundation Built

SQL rigor + Power BI interactivity + domain alignment

3

Next Steps

Advanced credit risk modeling, forecasting, portfolio optimization

Tools Used

SQL (T-SQL), Power BI, DAX, Power Query, Excel

Best Practices

Industry-aligned methodology suitable for real-world banking environments